Evidence Weighted Research Gap Validation

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***Abstract*—With the increasing volume of the scientific literature, it is becoming difficult for researchers to manually analyze the collected literature and find research gaps. Large Language Models (LLMs) can help in the analysis and synthesis of scientific literature; however, relying on pretrained knowledge may produce outdated or unsupported information, and a gap that an automated system reports may be an artefact of an incomplete literature collection. This work presents a Retrieval-Augmented Generation (RAG) based framework that separates *candidate* research-gap identification from *evidence-weighted validation*. Literature is collected from Semantic Scholar and OpenAlex, parsed into section-aware chunks, and indexed for hybrid BM25 and dense retrieval with mandatory cross-encoder reranking. A per-run knowledge graph supports four deterministic gap signals: explicit limitation statements, sparse concept combinations, temporal staleness, and sparse topic clusters. These are fused into a transparent candidate score and checked by a grounding gate; LLM-proposed candidates are optional and never decide that a gap exists. The Evidence-Weighted Research Gap Validation (EWRGV) stage then searches the run corpus and external bibliographic sources for direct, terminological, methodological, and counter-evidence, classifies the retrieved evidence with deterministic rules, measures coverage, and assigns a capped confidence, a classification (valid, uncertain, or unsupported), a reason code, and an evidence-chain explanation. The framework is implemented as a FastAPI service with persistent storage and a browser interface. The retrieval components were evaluated on the public TREC-COVID and SciFact benchmarks. Cross-encoder reranking improved TREC-COVID P@10 from 0.622 to 0.666 and nDCG@10 from 0.542 to 0.585. On SciFact evidence retrieval, BM25 (nDCG@10 0.804) outperformed the dense and hybrid configurations. Gap-identification and three-way validation accuracy were not measured, because no public dataset labels research-gap status; an evaluation protocol for them is defined.**

***Index Terms*—Retrieval-Augmented Generation, Large Language Models, Scientific Literature Analysis, Research Gap Identification, Knowledge Graphs, Evidence Validation, Natural Language Processing**

I. Introduction

The rapid growth of scientific publications has made it increasingly difficult for researchers to manually analyze large volumes of scientific literature. To understand the current state of a research field, researchers often need to study multiple papers, compare different approaches, identify their limitations, and explore possible directions for future work. As the amount of available literature continues to increase, performing these tasks manually becomes increasingly time-consuming and demanding.

Large Language Models (LLMs) have shown strong capabilities in tasks such as summarization, information extraction, and text generation. However, conventional LLMs rely on knowledge acquired during training and may therefore provide outdated or unsupported information when used for research-related tasks. Retrieval-Augmented Generation (RAG) addresses this limitation by combining information retrieval with language-model generation, allowing relevant external information to be incorporated into the generated response [1], [2].

RAG-based approaches have increasingly been applied to scientific literature analysis, including agent-based question answering over papers [3], citation-graph retrieval [4], and citation-aware synthesis [5], [6]. Several studies have explored automated research-gap identification using natural language processing, semantic clustering, embeddings, and LLMs [7]–[10], and recent systems combine hybrid retrieval, knowledge graphs, and gap scoring [11], [12].

However, automatically identifying a potential research gap does not necessarily mean that the gap represents a genuine or unresolved research problem. An apparently underexplored area may result from incomplete literature coverage, from terminology differences between communities, or from recent studies that have already addressed the issue. The reliability of an identified gap therefore depends on the completeness of the retrieved literature, the inclusion of recent studies, and the availability of evidence both for and against the claim.

Motivated by this need, this work presents a RAG-based framework for research paper analysis and automated research-gap identification whose central component, Evidence-Weighted Research Gap Validation (EWRGV), is kept separate from candidate generation. The objectives are (i) to construct a traceable, section-aware literature corpus for a research question; (ii) to identify candidate gaps from explicit and structural evidence in that corpus; and (iii) to assess each candidate against supporting and counter-evidence gathered beyond the initial retrieval. The work is guided by three research questions:

- **RQ1:** Can candidate gaps be produced from deterministic, corpus-derived signals so that every candidate is

traceable to paper-, section-, and page-level evidence?
- **RQ2:** Can candidate identification be separated from a validation stage that actively searches for counter-evidence outside the collected corpus?
- **RQ3:** How can a gap verdict expose its uncertainty when literature coverage is incomplete?

The contributions of this paper are:

1) an implemented pipeline in which gap identification uses four deterministic signals over a per-run knowledge graph with an interpretable GapScore and a grounding gate, while LLM proposals remain optional and subject to the same validation;
2) an implemented validation stage (EWRGV) that searches the run corpus, OpenAlex, and Semantic Scholar for direct, terminological, method-variation, and counter-evidence, and that bounds confidence when evidence or corpus adequacy is insufficient;
3) an explicit separation of what each stage can and cannot establish (Table II); and
4) a benchmark evaluation of the retrieval components on TREC-COVID and SciFact, and a protocol for evaluating gap identification and validation.

The combination of these elements, rather than any single technique, is the contribution: hybrid retrieval, knowledge graphs, gap scoring, and limitation extraction each appear in prior work (Section II). No empirical superiority is claimed.

II. Literature Survey

This survey discusses fifteen works selected from a verified reading list as the most directly relevant to RAG-based literature analysis, research-gap identification, and evidence validation. Unless marked otherwise, the statements below are based on the published abstracts; summaries of the wider reading list are provided in the supplementary review.

A. RAG for Scientific Literature Analysis

RAG combines a parametric generator with a non-parametric retrieval memory; Lewis et al. retrieved Wikipedia passages with a dense retriever and conditioned a sequence-to-sequence generator on them, and reported more specific and factual generation than a parametric-only baseline [1]. A survey of RAG for LLMs organises the field into naive, advanced, and modular RAG and identifies hallucination, outdated knowledge, and untraceable reasoning as the problems RAG addresses [2]. PaperQA introduced an agent that searches full-text papers, gathers and summarises evidence, and answers with citations; its authors report no hallucinated citations in their tests and note that the system assumes the underlying papers are correct [3]. CG-RAG integrates sparse and dense retrieval signals within citation-graph structures for research question answering [4], and SciRAG combines adaptive retrieval, citation-graph reasoning, and outline-guided synthesis [5]. OpenScholar retrieves from 45 million open-access papers and reports citation accuracy on par with human experts on its benchmark [6]. These systems address question answering and synthesis rather than deciding whether an unresolved problem exists.

B. Automated Research-Gap and Limitation Identification

GapFinder applies NLP to PDF documents to map potential research gaps [7]. GapSpotter uses semantic clustering with LLM agents to categorise gaps as methodological, theoretical, data-based, or population-based [8]. Frankowski et al. score peripheral keyword clusters using z-score peripherality, TF-IDF, rarity, and centroid distance, and validate the identified gaps with LLM annotators [9]. LLM-based literature scoping has also been applied to gap identification in clinical research [10]. LimitGen evaluates whether LLMs can identify the limitations of research papers and reports that literature retrieval grounds this ability [13]. HySemRAG combines hybrid retrieval, knowledge-graph traversal, agentic self-correction, and citation verification for methodological gap analysis [11]. ResearchRAG combines an academic knowledge graph, a graph-attention module that predicts knowledge density to locate under-explored concept clusters, and a citation-aware self-correcting generation loop [12]; it is a preprint that has not been peer reviewed, and its density-based gap score is computed in this work as a comparison baseline.

C. Evidence Verification

Scientific claim verification retrieves abstracts and labels them as supporting or refuting a claim, with rationales [14]. This task is the closest analogue to the classification of validation evidence in the present work; here, however, the claim being checked is a gap statement, which asserts the *absence* of research.

D. Research Gap Derived from the Literature

Table I compares the reviewed systems, and three observations follow. First, scientific RAG systems [3]–[6] provide retrieval and citation grounding, but they target answers and syntheses rather than the existence of an unresolved problem. Second, gap-identification approaches [7]–[9], [12] infer gaps from what is sparse or peripheral within the collected literature, so their output depends on that collection being complete. OpenScholar's datastore, for example, consists of open-access papers only [6]. Third, verification work [14] shows that claims can be checked against retrieved evidence, but it has been applied to positive claims rather than to gap statements. Our interpretation, which remains to be tested empirically, is that a gap statement should be treated as a claim of absence and checked against evidence gathered beyond the corpus that produced it. The present framework is designed around this interpretation.

III. Proposed Methodology

A. System Overview

The framework is a staged RAG pipeline for analysing scientific literature and identifying potential research gaps. Its central design principle is the separation between *identifying* a candidate gap and *validating* it. As shown in Fig. 1,

TABLE I
COMPARISON OF CLOSELY RELATED SYSTEMS (BASED ON PUBLISHED ABSTRACTS; "–" = NOT REPORTED IN THE ABSTRACT)

Work	Retrieval / evidence source	Structure used	Gap or limitation output	Check of the output against further evidence
PaperQA [3]	Agentic search over full-text papers	–	Answers (not gaps)	Citation-grounded answer
CG-RAG [4]	Sparse + dense signals in graph retrieval	Citation graph	Answers (not gaps)	–
SciRAG [5]	Adaptive sequential/parallel retrieval	Citation graph	Syntheses (not gaps)	Outline critique and refinement
OpenScholar [6]	45M open-access papers	–	Syntheses (not gaps)	Self-feedback inference loop
GapFinder [7]	PDF text mining	–	Potential gaps	–
GapSpotter [8]	Multiple papers on a topic	Semantic clusters	Categorised gaps	–
Frankowski et al. [9]	SLR records, keyword embeddings	Keyword clusters	Peripheral-cluster gaps	LLM annotators
LimitGen [13]	Literature retrieval for one paper	–	Limitations of a paper	Benchmark of human-written limitations
HySemRAG [11]	Semantic + keyword + KG traversal	Neo4j KG, topics	Methodological gaps	Agentic QA, citation verification
ResearchRAG [12]	KG-augmented retrieval	Academic KG, GNN density	Under-explored clusters	Self-correcting generation loop
This work	BM25 + dense, RRF, cross-encoder; OpenAlex, Semantic Scholar	Deterministic per-run KG	Candidate gaps from four signals (LLM optional)	Separate EWRGV stage: counter-evidence search, external coverage, capped confidence

TABLE II
WHAT EACH STAGE ESTABLISHES, AND WHAT IT DOES NOT

Stage	Establishes	Does not establish
Candidate identification (D1–D4, optional LLM)	A possible gap and the corpus signals behind it	That the gap is real or unresolved
GapScore and grounding gate	In-corpus support; statement–evidence agreement	Absence of work outside the corpus
EWRGV searches	Related, supporting, and counter-evidence in the corpus and in two external indices	Exhaustive coverage of the literature
Confidence and classification	Credibility relative to the searched evidence	Proof that the gap is genuinely unresolved

a research question is analysed and expanded, literature is collected and processed into a corpus, a knowledge graph is built over the corpus, and candidate gaps are identified, scored, grounded, and ranked. Selected candidates are then validated by EWRGV, which searches for supporting and counter-evidence, analyses coverage, and assigns a confidence, a classification, and an evidence-chain explanation. Table II states what each stage can and cannot establish.

B. Research Question and Query Expansion

The process begins with a research question. When an LLM provider is configured, a query-understanding step returns a structured representation of the question (main concepts, methods, datasets, terminology and method variants, query type) and up to eight expanded queries. The expanded queries account for differences in terminology across papers and also seed the knowledge graph and the later terminology searches of EWRGV. Without an LLM key, the question is stored without expansions.

C. Literature Collection and Document Processing

The original question and at most three expansions are sent to Semantic Scholar and OpenAlex, with at most 20 results per query and provider and 200 results in total. Records are deduplicated by DOI and normalised title, and open-access full texts are located through Unpaywall and provider links. PDFs are parsed page by page with PyMuPDF (no OCR). Headings are detected and classified into section types (e.g., abstract, methods, results, limitations, future work), and sections are split into chunks of about 500 tokens (maximum 600, overlap 75). Each chunk retains its paper identifier, section title and type, and page range, so all later evidence remains traceable to its source.

D. Knowledge Graph Construction

For each run, a knowledge graph is built from the corpus. (i) A relevance gate compares each paper's title and abstract with the centroid of the question and its expansions; papers below $\max(0.55,\ \tilde{s} - 0.08)$, where $\tilde{s}$ is the run median, are marked off-topic and excluded from detection. (ii) Entities of eight types (method, task, metric, modality, disease, dataset, population, concept) are tagged using seed lexicons, query-understanding terms, and patterns. (iii) Entity names of the same type whose embeddings have cosine similarity ≥ 0.88 are merged, and all surface forms are kept as aliases. (iv) Gap statements are extracted from sentences containing cue phrases of three classes, open problem (strength 1.0), limitation (0.8), and future work (0.6), weighted by section type. (v) Paper-

level co-occurrence is scored by pointwise mutual information, and concept communities are found with the Louvain method.

E. Hybrid Retrieval and Reranking

Chunks are retrieved with BM25 and with dense cosine similarity over stored chunk embeddings. The two ranked lists are fused by Reciprocal Rank Fusion, $\mathrm{RRF}(d) = \sum_i 1/(k + r_i(d))$ with $k = 60$ [15], and at most three hits per paper are kept. The fused candidates are then rescored by a cross-encoder (`ms-marco-MiniLM-L-6-v2`). Reranking is mandatory for the retrieval, gap-detection, and validation paths: if the model cannot be loaded, the request fails rather than silently returning unreranked results. For gap evidence, retrieval can be restricted to limitation, future-work, discussion, conclusion, and results sections.

F. Candidate Research Gap Identification

Four detectors operate on the knowledge graph and corpus. Their outputs are *candidates*, not accepted gaps.

- **D1, explicit statements.** Gap statements are embedded and clustered by average-linkage agglomerative clustering (cosine ≥ 0.72). A cluster stated by several papers forms one candidate with support $S = \sum_p \max_{s \in p} \mathrm{strength}(s)$ and $E = 1 - e^{-S/2}$. Later papers whose abstract, methods, or results mention all of its core concepts are reported as early counter-hints.
- **D2, combination sparsity.** For entity pairs of complementary types that each appear in at least three papers, the lift $\ell = n_{AB}/(n_A n_B/N)$ is computed. Pairs with $\ell \leq 0.35$ whose neighbourhoods in the co-occurrence graph overlap (Jaccard ≥ 0.25, a link-prediction signal) become candidates with sparsity $1 - \min(\ell, 1)$.
- **D3, temporal staleness.** For each concept, internal staleness combines the years since its last mention with the slope of its yearly share. External staleness uses OpenAlex yearly counts of the concept within the field. Concepts with staleness $\tau \geq 0.65$ in at least three papers become candidates.
- **D4, topic density.** Chunk embeddings are reduced to ten dimensions by PCA and clustered with HDBSCAN. When the clusters degenerate (for example, by paper), Louvain concept clusters are used instead. Topics with density $\rho \leq 0.35$, at least two papers, and relevance to the question become candidates; topic novelty η is also computed.

Optionally, an LLM can propose additional candidates from reranked passages. A proposal is kept only if it cites passage identifiers present in its input, and it is scored before validation by an evidence scorer based on lexical relevance, section quality, recency, paper independence, and a contradiction penalty. An LLM can also reword signal candidates, but the rewording is kept only if it passes the grounding gate. Both options are disabled by default, and the source of each candidate (signal, LLM, or both) is recorded.

G. Candidate Scoring, Grounding, and Ranking

Detector outputs describing the same gap are fused. The GapScore is a weighted mean of the available terms, explicit support E (0.25), sparsity S (0.25), staleness τ (0.15), novelty η (0.10), and paper independence $I = \min(n_{\text{papers}}/3, 1)$ (0.15), plus an agreement bonus $0.10 \cdot (n_{\text{detectors}} - 1)/3$, capped at 1. A contradiction term (0.10) is defined but not yet implemented and is therefore excluded from the weights. Sparsity is discounted by $\min(1, N_{\text{on-topic}}/50)$ because sparsity in a small corpus is weak evidence. The ResearchRAG density score $0.45(1 - \rho) + 0.35\tau + 0.20\eta$ [12] is computed alongside as a baseline ranking.

Each candidate then passes a grounding gate. A candidate with stated support must reach a cosine similarity of at least 0.60 between its statement and its evidence, with up to three re-retrieval attempts. A purely structural candidate (D2–D4) is grounded when evidence passages exist for its concepts or topic. Candidates that fail are marked *uncertain* but retained. Candidates are ranked greedily by GapScore with a small penalty for repeating a gap type (0.05 per earlier candidate of the same type, at most two). By default, validation processes the first ten candidates, interleaving signal and LLM candidates when both exist.

H. Evidence-Weighted Research Gap Validation

EWRGV assesses whether a candidate remains credible once additional evidence is considered; it does not treat the absence of an initially retrieved paper as evidence that a gap exists. Verification queries are generated from the candidate's concepts and their knowledge-graph neighbourhood, without an LLM:

- *direct search* combines the concepts with a field anchor;
- *terminology search* replaces concepts with graph aliases and synonyms;
- *method-variation search* substitutes sibling concepts of the same type; and
- *counter-evidence search* phrases the query as if the gap had been solved.

Each query is run over the run corpus (with hybrid retrieval and reranking) and over OpenAlex and Semantic Scholar (titles and abstracts). Sources that do not respond are recorded as unavailable instead of causing failure. While fewer than five relevant items are found, retrieval coverage is below 0.5, or not all concepts are covered, broader refined queries are issued, up to three iterations.

I. Evidence Classification and Coverage Analysis

Each retrieved item is classified by deterministic rules that combine its embedding similarity to the gap statement (r) with the share of gap concepts it names (c). Items with $r < 0.45$ are *insufficient*. An item stating that the topic has been widely studied with $c \geq 0.5$ is *contradictory*. An item reporting a solution that names all concepts with $r \geq 0.62$ is *counter-evidence*, and one with partial coverage is *partial*. An item containing a limitation cue with $c \geq 0.5$ or $r \geq 0.55$ is *supporting*. Evidence strength is r multiplied by a recency

factor (1.0 if the item is newer than the gap's latest statement, otherwise 0.8) and a source factor (1.0 for corpus full text, 0.9 for external abstracts). Duplicates found by several searches are merged by DOI, paper, or title.

Coverage is measured along five dimensions: retrieval, concept, terminology, and method coverage (the share of queries or concepts with relevant hits) and corpus adequacy, $0.5\min(1, n_{\text{run}}/10) + 0.5\min(1, n_{\text{ext}}/15)$. External bibliometric coverage, $\text{cov}_{\text{ext}} = n(C_1 \wedge \cdots \wedge C_k \wedge F)/\min_i n(C_i \wedge F)$, is computed from OpenAlex counts within field F. It identifies gaps that are sparse in the run corpus but well covered in the wider literature.

J. Confidence Assessment, Classification, and Explanation

Supporting mass $s = 2\,\text{GapScore} + \sum \text{str}(\text{supporting})$ is compared with counter mass $c = 1.2\sum \text{str}(\text{counter}) + 0.6\sum \text{str}(\text{partial}) + 1.2\sum \text{str}(\text{contradictory})$, giving $b = s/(s+c)$. Confidence is $b\,(1-0.6\,p)$, where $p = \text{clip}((\text{cov}_{\text{ext}} - 0.02)/0.13, 0, 1)$. Two caps implement the principle that sparsity is not proof of a gap. With fewer than three relevant items, or retrieval coverage below 0.25, confidence is capped at 0.39, so the candidate cannot be classified as valid or uncertain. With corpus adequacy below 0.4, it is capped at 0.69, so the candidate cannot be classified as valid. The resulting classifications are VALID_GAP (≥ 0.70), UNCERTAIN_GAP (≥ 0.40), and UNSUPPORTED_GAP. A reason code (supported, addressed externally, corpus artefact, contested, insufficient evidence, or weak support) and a template-based evidence-chain explanation listing queries, evidence, coverage, and scoring accompany each verdict. All weights and thresholds are design choices that have not yet been calibrated against labelled data.

IV. Block Diagram / Framework

Fig. 1 shows the complete workflow as implemented. The stages are exposed as separate REST endpoints that run as background jobs and are executed in order by the user or the interface; the API rejects a stage whose persisted prerequisites are missing. Solid arrows denote the default path and dashed arrows the optional LLM paths. Fig. 2 details the validation stage, including the iteration loop.

The components interact as follows. Lane A produces chunk records carrying paper, section, and page provenance, together with embeddings and a BM25 index. Lane B reads the corpus to build the knowledge graph, runs the detectors, and persists the ranked candidates with their signals, score breakdown, supporting statements, evidence passages, and counter-hints. Lane C reads these candidates and the graph, issues verification queries, and persists for each candidate its classification, confidence, coverage, evidence list, queries, and explanation. The interface reads the persisted results for display.

V. Algorithms and Technology Used

Algorithms 1 and 2 summarise candidate identification and validation. Table III lists the technologies used and the status of each.

Algorithm 1 Signal-based candidate gap identification

Require: knowledge graph G, run corpus $\mathcal{C}$, queries Q
$R \leftarrow \text{HybridRerank}(Q, \mathcal{C}$, gap-evidence sections)
$\mathcal{K} \leftarrow$ D1(G)$\cup$ D2(G)$\cup$ D4($G, \mathcal{C}, Q$)$\cup$ D3(G, OpenAlex trends)
$\mathcal{F} \leftarrow \text{Fuse}(\mathcal{K})$ ▷ merge candidates on shared concepts/evidence
for all $g \in \mathcal{F}$ **do**
  $g.score \leftarrow \text{GapScore}(E, S, \tau, \eta, I)$ + agreement bonus
  $\text{Ground}(g, R)$ ▷ cos ≥ 0.60 or mark uncertain
end for
optional: add grounded LLM proposals; LLM rewording kept only if grounded
return $\text{DiversityRank}(\mathcal{F})$

Algorithm 2 EWRGV validation of one candidate gap

Require: candidate g, graph G, sources {corpus, OpenAlex, S2}
$Q \leftarrow \text{VerificationQueries}(g, G)$; $H \leftarrow \emptyset$; $i \leftarrow 0$
repeat
  $i \leftarrow i + 1$; $H \leftarrow H \cup \text{Search}(Q)$ over all sources
  $E \leftarrow \text{Aggregate}(\text{Classify}(g, H))$ with strengths
  $cov \leftarrow \text{Coverage}(E, Q)$; $Q \leftarrow \text{Refine}(g, G)$
until $i = 3$ **or** ($|E_{\text{relevant}}| \geq 5$ **and** retrieval cov. ≥ 0.5 **and** concept cov. $= 1$)
$cov_{\text{ext}} \leftarrow \text{ExternalCoverage}(g$, OpenAlex)
$\kappa \leftarrow \text{Confidence}(g.score, E, cov, cov_{\text{ext}})$ with caps
return class(κ), reason code, evidence chain

TABLE III
Technologies and algorithms used

Component	Technique / library (purpose)	Status
Query understanding	LLM via OpenRouter/OpenAI-compatible API, temperature 0 (concepts, expansions)	Impl.
Collection	Semantic Scholar, OpenAlex APIs; Unpaywall (open-access PDFs)	Impl.
Parsing	PyMuPDF page text; rule-based heading detection; no OCR	Impl.
Sparse retrieval	BM25 (`rank_bm25`)	Impl.
Dense retrieval	Sentence embeddings; exact cosine search	Impl.
Fusion	Reciprocal Rank Fusion, $k = 60$ [15]	Impl.
Reranking	Cross-encoder `ms-marco-MiniLM-L-6-v2` via sentence-transformers; fail-closed	Impl.
Graph analysis	PMI co-occurrence; Louvain	Impl.
Clustering	Agglomerative (D1); PCA + HDBSCAN (D4)	Impl.
LLM candidates	One structured LLM call; cited-ID grounding check	Optional
Evidence classifier	Embedding similarity + cue rules	Impl.
LLM evidence classifier	Interface only	Not impl.
Contradiction signal	Weight reserved in GapScore	Not impl.
Service	FastAPI, background jobs; PostgreSQL/Supabase, pgvector, JSONB; object storage	Impl.

The methods were chosen for specific reasons. BM25 and dense retrieval are complementary: BM25 matches exact terminology, while dense retrieval captures paraphrases. RRF fuses their rankings without calibrating incompatible scores [15]. Cross-encoder reranking scores each query–passage pair jointly to improve precision at the top of the fused list. Detection and validation are deterministic for a fixed corpus, embedding model, and external-source state, which makes every verdict reproducible and inspectable. LLMs are con-

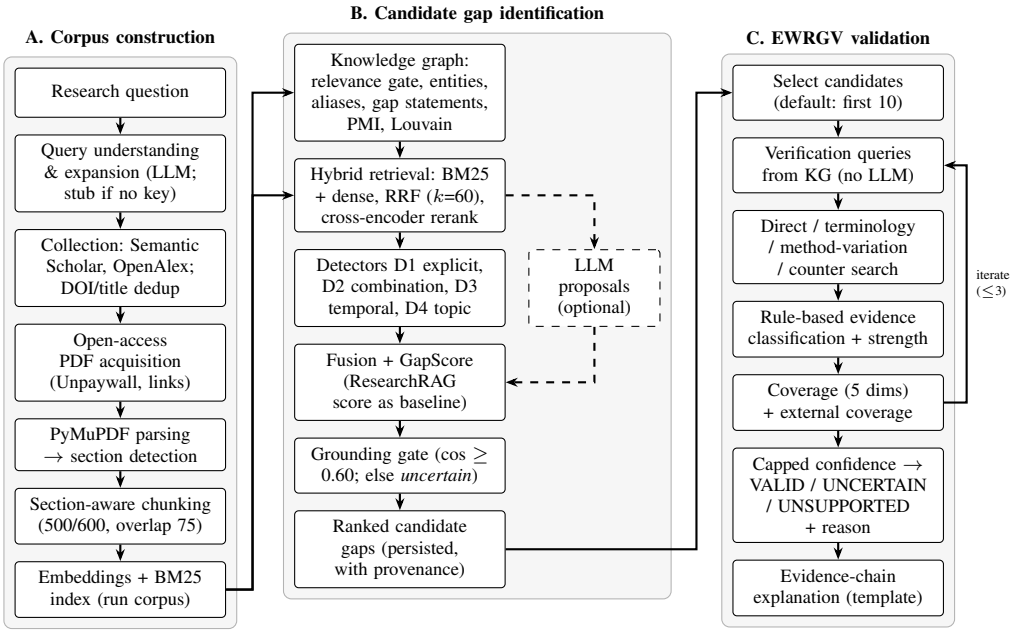


Fig. 1. Overall workflow of the implemented framework. Lane A builds a traceable run corpus, lane B identifies and ranks candidate gaps (dashed: optional LLM path), and lane C validates selected candidates with EWRGV. Each stage is a separate API job executed in order.

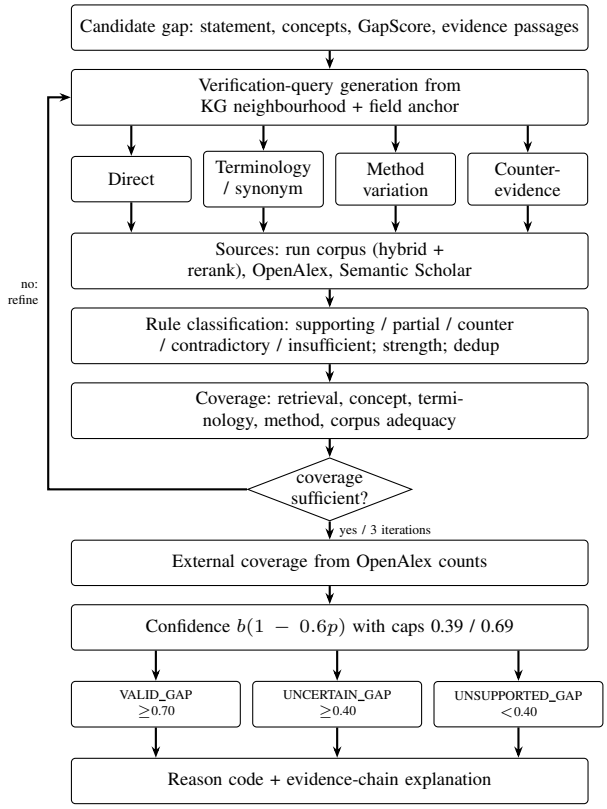


Fig. 2. Evidence-Weighted Research Gap Validation (EWRGV) as implemented: deterministic query generation, four search directions over three sources, rule-based classification, coverage-driven iteration, external coverage, capped confidence, and explained classification.

fined to query understanding and optional, grounding-checked proposals. The embedding model is configurable: the default configuration uses an OpenAI embedding model (1536 dimensions), while a database migration targets the local model `bge-small-en-v1.5` (384 dimensions). The model used for a given run must be reported with its results; the benchmark evaluation in Section IX used `bge-small-en-v1.5`.

VI. Dataset Description

A. Per-Run Literature Corpus

The framework does not use a fixed benchmark dataset. Each research question produces its own run corpus through the collection and processing stages of Section III. Table IV summarises how a corpus is constructed. The selection criteria are relevance to the question and its expansions (as ranked by the search providers), the per-query and total limits, deduplication, and the relevance gate of the knowledge graph. Off-topic papers are retained in the corpus but excluded from gap detection.

Run-corpus statistics. No verified export of a run's corpus, candidates, or validations was available when this paper was prepared, so run-corpus sizes and distributions are not reported.

Limitations. Full text is limited to open-access papers, and the remaining records contribute only metadata. Scanned PDFs yield no text because OCR is not used. The two search providers rank by their own relevance models, and the result caps bound recall. The entity lexicons include domain packs for diseases and imaging modalities, so entity tagging is likely to be strongest in biomedical imaging and needs extension for other fields. External validation evidence consists of titles and abstracts only.

TABLE IV
RUN-CORPUS CONSTRUCTION (CONFIGURATION DEFAULTS)

Aspect	Setting
Sources	Semantic Scholar, OpenAlex (metadata); Unpaywall and provider links (PDFs)
Queries	Question + up to 3 expansions
Limits	20 results per query per provider; 200 in total
Deduplication	DOI, then normalised title
Full text	Open-access PDFs only; text layer only (no OCR)
Units	Section-typed chunks, ≈500 tokens, overlap 75, with page range
Relevance gate	Paper vs. question centroid, $\max(0.55, \tilde{s} - 0.08)$
External validation evidence	OpenAlex and Semantic Scholar titles/abstracts, 10 results per query

TABLE V
PUBLIC BENCHMARK DATASETS USED

	TREC-COVID (Round 5)	SciFact (dev)
Documents	191,175 CORD-19 IDs (title + abstract)	5,183 abstracts
Queries	50 topics (query field)	188 evidence-bearing claims
Labels	69,318 graded judgements (0–2)	Evidence documents (binary; stance ignored)
Task	Ad-hoc scientific retrieval	Claim-evidence retrieval

B. Public Benchmark Datasets

Two public benchmarks with relevance labels were used to evaluate the retrieval components (Table V).

- **TREC-COVID**[1] (final Round 5) provides graded topic–document judgements over the CORD-19 release of 2020-07-16. The 191,175 valid document IDs were loaded from the official metadata, using title and abstract only (191,132 have usable text). The collection has 50 topics and 69,318 judgements (grades 0–2); 26,664 judgements have a grade above 0.
- **SciFact** [14][2] provides 5,183 abstracts and 300 development claims. Of these claims, 188 carry evidence annotations (124 SUPPORT, 64 CONTRADICT) referencing 182 distinct documents; the 112 claims without evidence are excluded from retrieval scoring.

Only the SciFact development split was used, and no training data informed inference. CORD-19 records carry individual licences; SciFact claims are CC BY 4.0 and its abstracts ODC-By 1.0. The data were used locally and are not redistributed.

TREC-COVID tests ad-hoc scientific retrieval with graded labels. SciFact tests the retrieval of evidence for scientific claims, which is the closest public proxy for EWRGV's evidence search. *Neither dataset labels research gaps or EWRGV's three validation classes.*

VII. PERFORMANCE METRICS

Retrieval is evaluated with standard ranking metrics on the labelled benchmarks, and gap identification and validation with classification metrics against human labels (Table VI).

[1]https://ir.nist.gov/trec-covid/data.html

[2]https://github.com/allenai/scifact

TABLE VI
EVALUATION METRICS AND THEIR STATUS

Metric	Measures	Status
P@5, P@10	Early precision of the ranking	Measured
R@10	Share of all relevant documents found in the top 10	Measured
MRR	Rank of the first relevant document	Measured
MAP	Precision over all relevant documents	Measured
nDCG@10	Graded quality of the top 10	Measured
judged / unjudged@10	Label coverage of retrieved slots	Reported
Gap identification P / R / F1	Agreement of candidates with human gap labels	Not measured
Validation accuracy, macro-F1	Agreement of the three classes with adjudicated labels	Not measured
Agreement (κ), calibration	Label reliability; reliability of confidence	Not measured

Retrieval metrics. A document is relevant if its label grade is greater than 0. For a query q with relevant set Rel_q and top-k ranking Top_k:

- $\mathrm{P@}k = |\mathrm{Rel}_q \cap \mathrm{Top}_k|/k$, with the denominator fixed at k;
- $\mathrm{R@}10 = |\mathrm{Rel}_q \cap \mathrm{Top}_{10}|/|\mathrm{Rel}_q|$;
- MRR is the reciprocal rank of the first relevant document;
- average precision is $\mathrm{AP} = \frac{1}{|\mathrm{Rel}_q|} \sum_i \mathrm{P@}i \cdot \mathrm{rel}(i)$, and MAP is its mean over queries;
- nDCG@10 uses the graded gain $2^g - 1$ and the discount $\log_2(i+1)$, normalised by the ideal ranking.

All metrics are macro-averaged over queries. Following the `trec_eval` convention, unjudged retrieved documents count as non-relevant (grade 0). Because pooled and evidence annotations are incomplete, this can penalise a system for retrieving relevant but unjudged documents. The number of judged and unjudged documents in the top-10 slots (judged@10 / unjudged@10) is therefore reported alongside each result.

Gap-level metrics. Gap identification is to be measured with precision, recall, and F1 against human-labelled gap statements. Validation is to be measured with three-way accuracy and macro-F1 against adjudicated VALID/UNCERTAIN/UNSUPPORTED labels, together with inter-annotator agreement and calibration of confidence. These require a frozen candidate set labelled by blinded reviewers under agreed operational definitions; no public dataset provides such labels.

VIII. IMPLEMENTATION

The system is implemented in Python as a modular monolith built on FastAPI. Module boundaries follow the research boundary between detection and validation (Table VII). Domain models and interfaces are separated from providers (LLM, embeddings, search) and storage, and components receive their dependencies through constructors.

Execution. The stages are invoked in the order `/query`, `/collect`, `/acquire`, `/parse`, `/sections`, `/chunks`, `/represent`, `/kg/build`, `/gaps`, `/validate`. Long-running stages run as in-process background jobs whose status

TABLE VII
IMPLEMENTED MODULES (REPOSITORY `APP/`)

Module	Responsibility
`query_understanding`	LLM question analysis, expansions
`ingestion`	Collection, dedup, acquisition, parsing, sectioning, chunking, embedding
`retrieval`	BM25, dense, RRF, cross-encoder, section filters
`knowledge_graph`	Graph build, gap statements, job orchestration
`gap_detection`	D1–D4, fusion, GapScore, grounding; LLM detector and evidence scorer
`validation`	Query generation, four searches, sources, iteration, validator
`evidence`, `coverage`	Classification, strength, aggregation; coverage dimensions
`scoring`, `explanation`	Confidence, classification, reason codes; evidence chain
`api`, `ui`	REST routes, job status; browser interface

can be polled. The analysis routes return HTTP 409 when persisted prerequisites are missing. The browser interface presents this sequence as a gated pipeline and displays candidates, evidence, and validation results.

Storage. Paper records, provenance, chunk embeddings (pgvector), the knowledge graph, candidates, and validations (JSONB) are stored in PostgreSQL (Supabase), and PDFs and processing artefacts are kept in object storage. External search responses are cached so that re-running validation does not repeat API calls.

Robustness. External requests are rate-limited, retried with exponential back-off, and cached. An unavailable source is recorded in the validation output, and the reranker fails closed.

Incomplete or partial functionality. (i) There is no durable run coordinator: jobs run in the API process and are not resumed after a restart. (ii) The contradiction signal and the LLM evidence classifier are not implemented. (iii) OCR is not supported. (iv) The embedding model and vector dimension must be configured consistently with the database migration. (v) The interface has not been evaluated with users.

IX. EXPERIMENTS AND RESULTS

A. Benchmark Setup

The retrieval components were evaluated in a benchmark run of 2026-10-20, using the production code and settings unchanged. The configurations were:

- **BM25:** `BM25Index` (BM25Okapi, $k_1 = 1.5$, $b = 0.75$).
- **Dense:** the local sentence-transformers model `BAAI/bge-small-en-v1.5` (384 dimensions; normalised vectors, so cosine similarity is the dot product).
- **Hybrid:** the production RRF ($k = 60$) over the top 100 dense and top 100 BM25 results.
- **Reranking:** the production cross-encoder `ms-marco-MiniLM-L-6-v2` (CPU, batch size 16), rescoring the leading 50 results and keeping retrieval order after that window.

The retrieval depth was 100. Benchmark adapters converted each document's title and abstract into the system's chunk objects. The models were run offline from local caches, and no paid API was used.

Two runs did not complete. Dense indexing of the full TREC-COVID collection did not finish within the available time, so dense and hybrid results are not available for TREC-COVID. Cross-encoder reranking was not completed for SciFact. These configurations are reported as *not measured*.

TABLE VIII
TREC-COVID ROUND 5 RETRIEVAL (50 TOPICS, MACRO AVERAGE)

Method	P@5	P@10	R@10	MRR	MAP	nDCG@10
BM25	0.640	0.622	0.014	0.821	0.062	0.542
BM25 + CE (top 50)	**0.700**	**0.666**	**0.015**	**0.834**	**0.065**	**0.585**
Dense	not measured (run incomplete)					
Hybrid RRF	not measured (requires dense run)					

judged@10 / unjudged@10 (of 500 slots): BM25 434/66; BM25 + CE 436/64. CE = cross-encoder.

TABLE IX
SCIFACT DEV EVIDENCE RETRIEVAL (188 CLAIMS, MACRO AVERAGE)

Method	P@5	P@10	R@10	MRR	MAP	nDCG@10
BM25	**0.182**	**0.097**	**0.908**	**0.777**	**0.770**	**0.804**
Dense (BGE)	0.054	0.028	0.263	0.248	0.243	0.249
Hybrid RRF	0.063	0.048	0.457	0.311	0.300	0.314
+ CE rerank	not measured (run not completed)					

judged@10 / unjudged@10 (of 1,880 slots): BM25 182/1,698; dense 53/1,827; hybrid 90/1,790.

B. TREC-COVID Retrieval

Table VIII reports the 50-topic macro averages. Reranking the BM25 top 50 raised P@5 from 0.640 to 0.700, P@10 from 0.622 to 0.666, and nDCG@10 from 0.542 to 0.585; MRR and MAP increased slightly. The topics have on average more than 500 relevant documents each, so R@10 is necessarily small.

C. SciFact Evidence Retrieval

Table IX reports the results for the 188 evidence-bearing claims over the full corpus of 5,183 abstracts. BM25 was clearly strongest (R@10 0.908, nDCG@10 0.804). The dense model scored much lower (R@10 0.263, nDCG@10 0.249). RRF fusion fell between the two (R@10 0.457, nDCG@10 0.314), that is, below BM25 alone. Each claim has few evidence documents, so P@10 cannot be much above 0.1.

D. Gap Identification and Validation

Gap-identification precision, recall, and F1, and EWRGV three-way accuracy and macro-F1, were **not measured**. TREC-COVID labels topic–document relevance, and SciFact labels whether evidence supports or contradicts a claim. Neither states whether a research gap is real, uncertain, or unsupported, and SciFact's claims without evidence do not correspond to unsupported gaps. SciFact stance labels were therefore not substituted for EWRGV classes, since that would change the task. Published gap-dataset candidates were examined, but their annotations, provenance, or licences could not be confirmed, or their labels were generated by a model; none was used as ground truth.

TABLE X
SOFTWARE VERIFICATION EVIDENCE

Check	Outcome
Benchmark metric tests	2 passed (ranking metrics, duplicates, zero-relevant queries, macro aggregation, unjudged counts)
Unit tests (`tests/unit`)	474 passed; 21 modules not collected in the local environment (missing dependencies)
Integration / end-to-end	Not run

E. Software Verification

Table X summarises the automated tests.

F. Remaining Experiments

The following experiments *have not been run*:

- **E1:** complete the dense, hybrid, and reranked configurations on both benchmarks, with pinned model revisions, and evaluate on full-text, section-aware chunks.
- **E2:** gap-identification P/R/F1 and κ for signal-only, LLM-only, and combined candidates, using a blinded, adjudicated expert sample.
- **E3:** ablation of D1–D4, and GapScore versus the ResearchRAG baseline ranking.
- **E4:** three-way validation accuracy, macro-F1, confusion matrices, and calibration, including gaps known to have been addressed after the corpus period.
- **E5:** sensitivity of the verdicts to corpus size and to disabling each external source.

Results published for other systems use different datasets, metrics, and set-ups, and are not compared with these figures.

X. INFERENCES

The measured results concern the retrieval components on title-and-abstract benchmarks. They do not measure the quality of the identified gaps or of the validation verdicts.

Reranking. On TREC-COVID, cross-encoder reranking of the BM25 candidates improved early precision and nDCG@10 (Table VIII). This is consistent with the design decision to make reranking mandatory. It comes from a single run without significance testing.

Dense retrieval and fusion. On SciFact, BM25 outperformed both the dense model and equal-weight RRF (Table IX). Fusing a much weaker ranking lowered quality below that of BM25 alone. Because the corpus searches of EWRGV use hybrid retrieval with this embedding model, the result indicates that the dense component and the fusion weighting should be re-examined, for example with a scientific-domain embedding model or a weighted fusion. Whether this transfers to EWRGV's verification queries over full-text chunks has not been tested. The deployed configuration, hybrid retrieval followed by reranking, was not measured on SciFact.

Label coverage. SciFact's top-10 slots were mostly unjudged, and TREC-COVID's partly so. Because unjudged documents are scored as non-relevant, the absolute values are conservative.

RQ1, traceability. Every signal candidate carries its detector signals, score breakdown, supporting sentences, and passages with paper, section, and page provenance, and optional LLM candidates must cite input passages. Whether the candidates are *useful* gaps remains to be measured (E2).

RQ2, separation. Validation is a separate module and job. It issues its own queries without an LLM and searches beyond the corpus that produced the candidate, so a corpus-sparse candidate can be marked as a corpus artefact. Whether this reduces false gaps in practice is to be tested in E4 and E5.

RQ3, uncertainty. Confidence is bounded by explicit caps when evidence is thin or corpus adequacy is low, and each verdict carries a reason code. Verdicts are thus relative to the searched literature. The weights and thresholds are uncalibrated, so confidence values should not yet be read as probabilities.

Limitations. Both benchmarks are biomedical and use titles and abstracts, so neither represents interdisciplinary, full-text gap discovery. The rule-based evidence classifier may mislabel paraphrased or hedged evidence. Results depend on third-party search APIs whose rankings and coverage change over time. Model weights were identified by name, without pinned revisions.

XI. CONCLUSION

This paper presented a RAG-based framework for research paper analysis and automated research-gap identification. The framework separates candidate identification from evidence-weighted validation. Candidates come from four deterministic signals over a per-run knowledge graph, with a transparent GapScore and a grounding gate; LLM proposals are optional. Each candidate is then validated against direct, terminological, method-variation, and counter-evidence from the run corpus and two external indices, with coverage-aware, capped confidence and an evidence-chain explanation.

A partial benchmark evaluation of the retrieval components produced two findings. On TREC-COVID, cross-encoder reranking improved P@10 from 0.622 to 0.666 and nDCG@10 from 0.542 to 0.585. On SciFact evidence retrieval, BM25 outperformed the dense and hybrid configurations. Gap-identification and validation accuracy could not be measured, because no public dataset labels research-gap status.

Future work will (i) complete the dense, hybrid, and reranked benchmark runs and re-examine the embedding model and fusion weighting; (ii) build an adjudicated expert benchmark for gap identification and three-way validation; (iii) calibrate the scoring weights and thresholds; (iv) implement the contradiction signal; and (v) add durable orchestration, OCR, and domain-general lexicons.

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